ADVANCED TECHNOLOGY PIONEERS 2026 · DCT ABU DHABI CHALLENGE

Air2Stay

A what-if simulator that estimates how flight changes into Abu Dhabi change hotel check-ins and guest nights

DEFAULT DEMO: +3 WEEKLY INDIA FLIGHTS IN DECEMBER

+731
hotel check-ins / month

+2,706
guest nights / month

vs. the historical baseline; estimates, not guarantees

DCT Abu Dhabi challenge statement: "How might we build a tool that predicts how flight changes affect hotel visits in Abu Dhabi?"

WHAT THE PACKAGE CONTAINS

Web prototype
4 pages · 33 departure countries · one standalone HTML file

Calculation audit workbook
11 sheets · 396 route baselines · 17,820 bridge rows

Technical guide & user manual
15 sections: method, demo script, glossary

Complete source code
12 files: Python, JavaScript, HTML/CSS, PowerShell

AI-tool declaration: prototype build used AI-assisted coding (OpenAI Codex) and the cover concept used OpenAI image generation, both under team direction; the team can explain every method, decision and result in its own words.

Team Air2Stay

Advanced Technology Pioneers 2026 – DCT Abu Dhabi sponsor track, flight-to-hotel demand simulator

TEAM MEMBER

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A four-person team of BSc Computer Science students entering the DCT Abu Dhabi sponsor track, building a working, explainable flight-to-hotel demand simulator rather than a static report.

Flight data and hotel data mean different things by “country”

117,608	33	130	59	45	12
flight rows	departure countries	departure cities	airlines	hotel nationalities	hotel-only nationalities

FLIGHT FILE · AVIATION TEAM

“Country” = where the aircraft departed. Holds flights, seats, filled seats, passengers, P2P, transfer, transit.

HOTEL FILE · HOTEL TEAM

“Country” = the guest’s passport nationality. Holds daily New Arrivals, Guests and Same-Day Guests.

WHY THE TWO CAN’T SIMPLY BE MATCHED

Traveller	Flight file says	Hotel file says
Australian passport holder flies from London	United Kingdom	Australia
UK passport holder flies from Delhi	India	United Kingdom
Indian resident returns home	India	No hotel check-in
Transfer passenger continuing to Europe	India	No Abu Dhabi check-in

The 12 hotel-only nationalities (no same-named departure country): Australia, Brazil, Czechia, Denmark, Finland, Mexico, Morocco, Norway, Pakistan, Romania, South Africa, Sweden. The other 33 names match exactly.

Both files are correct: they answer different questions. Air2Stay builds an estimated bridge between them instead of pretending the fields are the same.

Every number comes from six supplied files

Supplied file	Data rows	Coverage	What Air2Stay uses it for
flight_data.xlsx	117,608	Jan 2022 – Feb 2026	Seats, PAX, P2P, transfer, transit, load factor, weekly frequency → 396 country-month route baselines
data international_train.xlsx	58,622	Jan 2022 – Jul 2025	Daily New Arrivals, Guests, Same-Day Guests for 45 nationalities → check-ins, stays, bridge, validation
data international_test.xlsx	9,202	Aug 2025 – Feb 2026	Future-style rows; Guests target intentionally missing
data domestic_train.xlsx	1,308	Jan 2022 – Jul 2025	Separate domestic daily stay model
data domestic_test.xlsx	212	Aug 2025 – Feb 2026	Domestic rows needing predicted Guests
Data_Dictionary.pdf	—	8 pages	Field definitions only; supplies no forecast values

Check-ins

Hotel New Arrivals (column C)

Guest nights

Sum of daily Guests: one person over 3 nights counts 3 times

Average stay

Total Guests ÷ total New Arrivals

Common grain

2022 flights are monthly, later years daily, so everything is aggregated to month

The DCT challenge web page supplied the business problem and success criteria, not data. Flight columns used: A Date, B Departure Country, F Avg Weekly Frequency, M Load Factor, N Total P2P, O Total PAX, Q Total Seats, R Transfer, S Transit.

Route baselines per country × month: load factor = PAX ÷ seats; P2P share = P2P ÷ PAX; weeks = days ÷ 7; seats per flight = seats ÷ (weekly frequency × weeks); hotel conversion = New Arrivals ÷ P2P (training months, month-calibrated).

A five-step chain turns airline seats into hotel guest nights

1

Scheduled seats

baseline + extra flights × seats × weeks

298,700

2

Passengers

seats × load factor (88.1%)

263,155

3

P2P visitors

passengers × P2P share (65.4%)

172,103

4

Hotel check-ins

P2P × hotel conversion (42.5%)

73,144

5

Guest nights

check-ins × average stay (3.32)

242,838

INDIA, DECEMBER: BASELINE VS. SCENARIO

Output	Baseline	Scenario	Change
Scheduled seats	296,096	298,700	+2,604
Passengers	260,890	263,155	+2,265
P2P visitors	170,544	172,103	+1,559
Hotel check-ins	72,413	73,144	+731
Guest nights	240,131	242,838	+2,706

Added seats: 3 flights × 196 seats × (31 ÷ 7 = 4.4286 weeks) = 2,604

Baseline dials (from history): load factor 88.11%, P2P 65.37%, conversion 42.46%, stay 3.316 nights

Hotel conversion combines travel purpose and hotel choice, which the data don't show separately. It is one labelled, editable dial.

READ THE HEADLINE CAREFULLY

The live demo uses rounded dials (88.1%, 65.4%, 42.5%, 3.32) vs. baseline 88.11%, 65.37%, 42.46%, 3.316. The workbook's isolated flights-only test (full-precision dials) gives +637 check-ins and +2,112 guest nights. The rest of +731 / +2,706 comes from that rounding.

The Country Bridge splits demand across 45 nationalities

It doesn't change the total. It only divides the check-ins and guest nights that the chain already calculated.

HOW EACH WEIGHT IS BUILT (PER ORIGIN × MONTH)

Inputs: monthly P2P passengers per origin vs. New Arrivals per nationality, on a log(1 + value) scale

Combined: max(0, 0.70 × level correlation + 0.30 × change correlation)

Score: nationality share × (0.35 + 2 × combined²)

Direct-match prior: × 6 when names match (India→India), a preference, not a forced 100%

Season: score × seasonal factor^0.7, factor clipped to 0.35–2.5

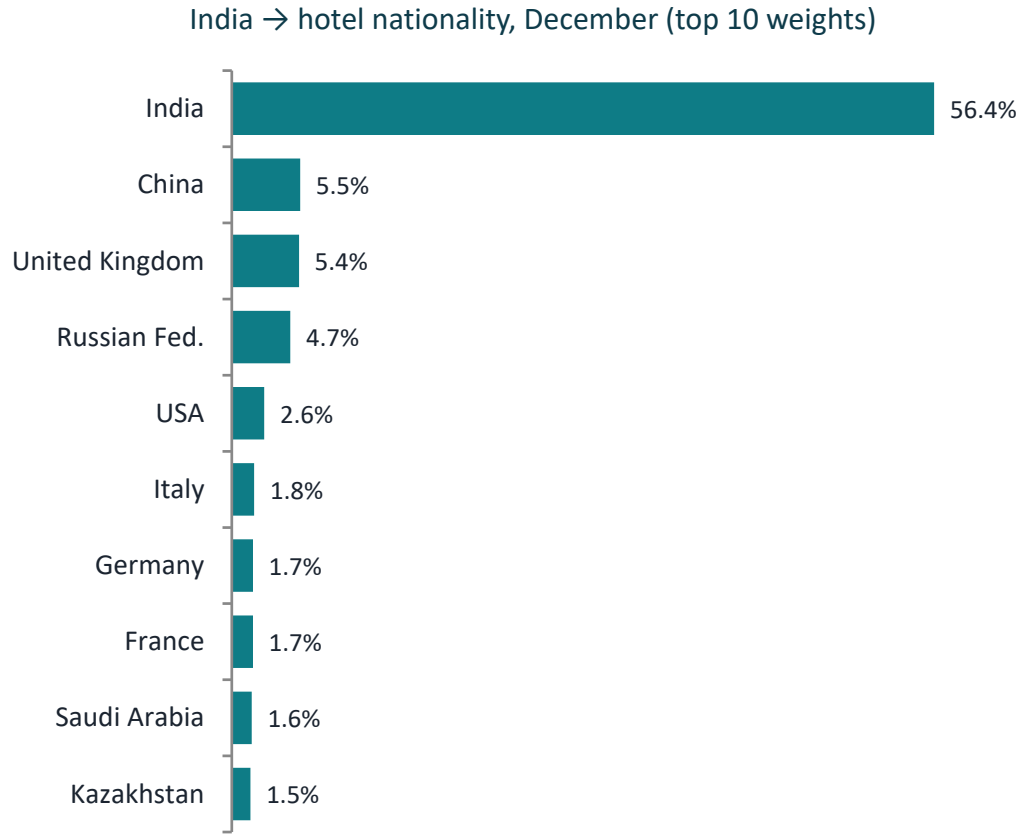
Normalise: weight = score ÷ sum of scores, so all weights add to 100%

OTHER RULES

Planner override: edit weights; each is divided by the new total so all 45 still add to 100%. Reset removes it.

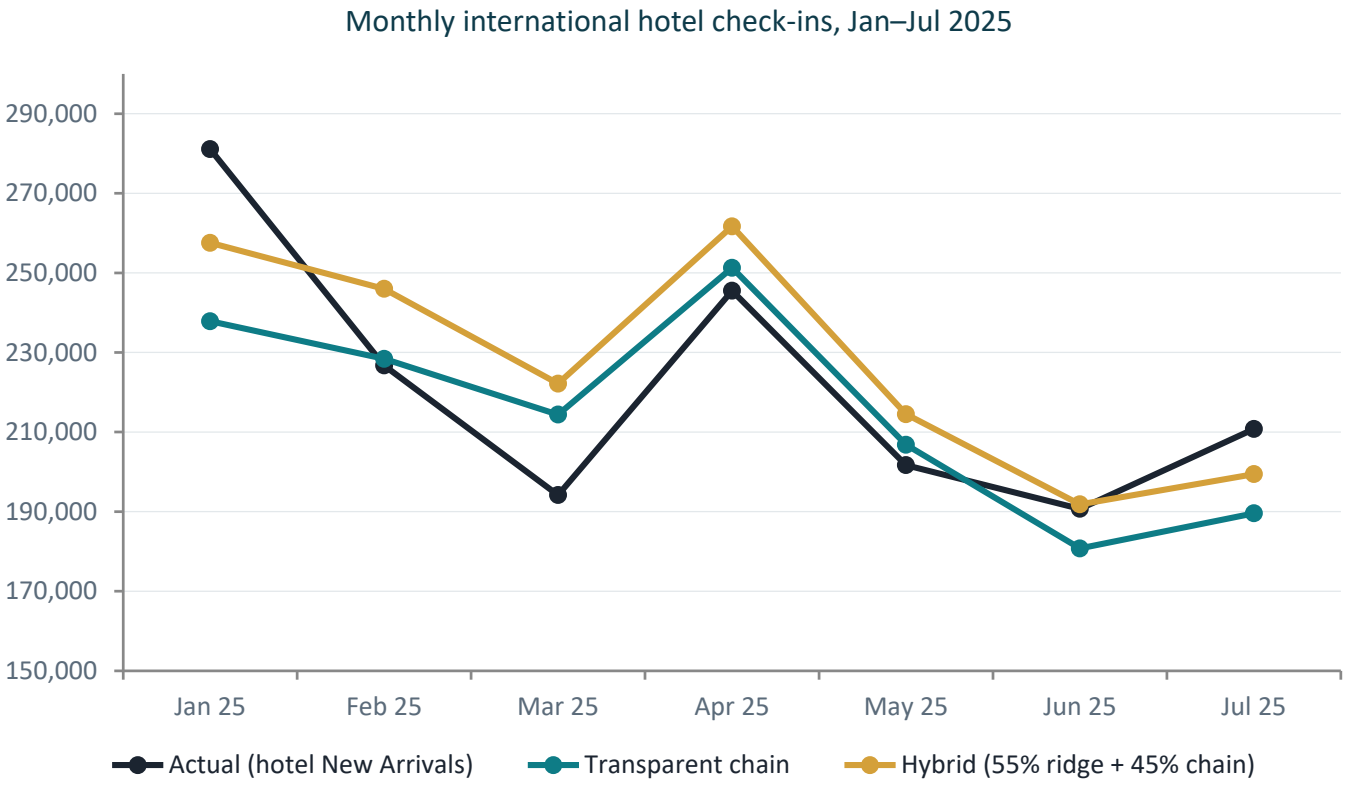
Per-nationality stay: 70% nationality-month + 30% nationality overall, clipped to 1–8 nights; rescaled so guest nights reconcile to the headline.

Not passenger tracking: an aggregate estimate, never for individual targeting.



Of +731 extra check-ins, ≈412.5 go to Indian nationals (56.4%). 17,820 rows = 33 origins × 12 months × 45 nationalities.

Tested on 7 later months it never trained on



Split by time: trained on 36 months (Jan 2022–Dec 2024, 83.7%), rules locked, then scored on 7 months (Jan–Jul 2025, 16.3%).

WMAPE = sum of |actual − predicted| ÷ sum of actual. Lower is better.

6.91%

Transparent chain

107,148 ÷ 1,550,679

7.24%

Hybrid benchmark

112,226 ÷ 1,550,679

11.28%

Daily intl. stay model

4,138 rows, May–Jul 2025

8.52%

Daily domestic stay model

separate domestic test

THE MACHINE-LEARNING LAYER

Ridge regression (regression with a stability penalty; features standardised, intercept not penalised)

Features: log P2P, log seats, load factor, P2P share, month sine/cosine, time trend

Decision: the hybrid scored worse, so the transparent chain is the engine. The hybrid is kept as evidence.

Ranges, sensitivity and data support show what to trust

P10 / P50 / P90, INDIA DECEMBER TOTAL GUEST NIGHTS

P10	P50	P90
219,185	**242,838**	**260,006**
× 0.9026 · cautious lane	× 1 · central formula result	× 1.0707 · higher lane

Multipliers are the 10th and 90th percentiles of actual ÷ hybrid ratios over the 7 validation months. P50 does not mean “50% accurate”. With only 7 months these are rough planning lanes, not a calibrated probability or 95% interval.

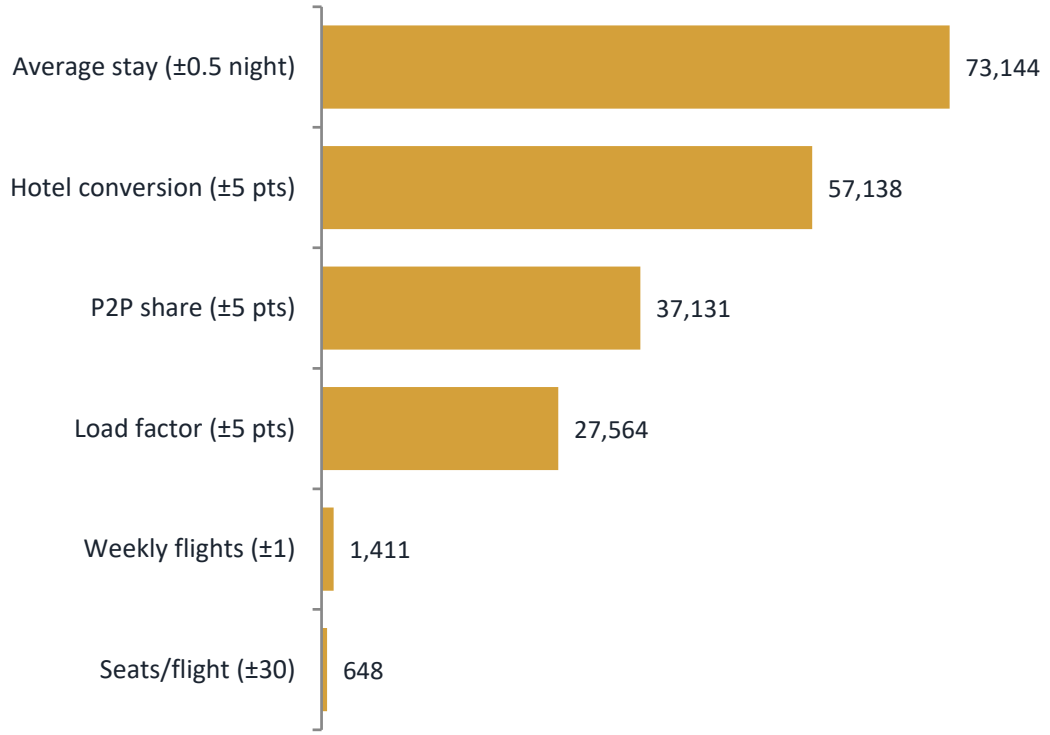
DATA SUPPORT BADGE (EVIDENCE COVERAGE, NOT CONFIDENCE)

Badge	Rule for the selected country + month
High	3 or more matching historical observations
Medium	1 or 2 matching observations
Low	None: fallback (zero baseline seats, overall averages, seasonal nationality mix)

INTENDED IMPACT

Shows DCT planners the likely hotel demand effect of a route change in advance, turning capacity planning into a shared, numbers-based conversation.

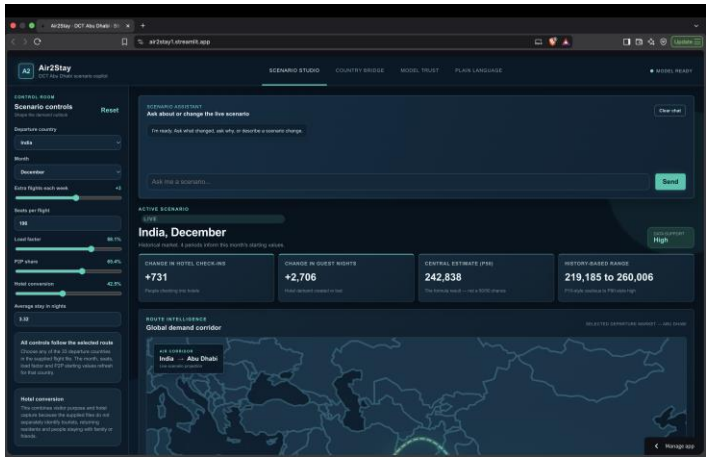
Sensitivity: guest-night span, low vs. high input



One input moves down and up while the others stay fixed. The biggest levers are stay length and hotel conversion, not seat count, so DCT should firm up those assumptions first. This is a decision guide, not a causal claim.

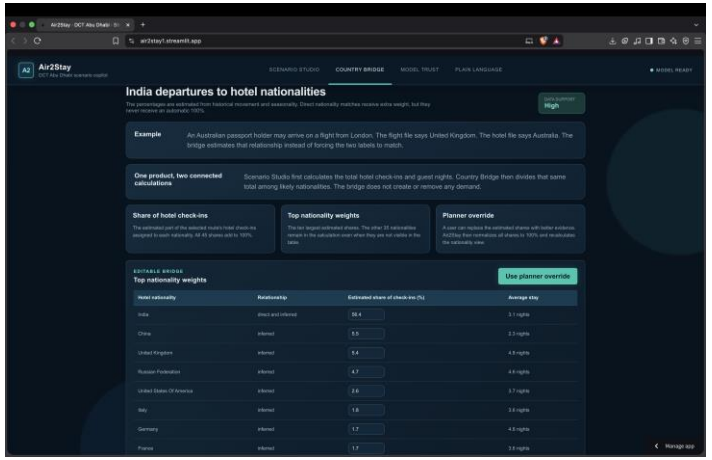
The working prototype, screen by screen

Live screenshots from the standalone web app — not mockups



Scenario Studio

Change dials, see check-ins and nights update live



Country Bridge

Splits the total across 45 likely nationalities



Model Trust

Actual vs. predicted, with both WMAPE scores

A fourth page, Plain Language, turns these same numbers into planner-ready sentences. One standalone HTML file, 33 departure countries, no server required.

Responsible use, known limits and the ask

PRIVACY & SAFETY

- Aggregate country-month data only: no names, passports, bookings or emails
- Standalone HTML runs locally; no external API receives the data
- Overrides change the browser session only; they should cite a named source (not stored in the prototype)
- Outputs are discussion prompts; a human approves decisions
- 83.7/16.3 time split stops data leakage. It is model governance, not cybersecurity

KNOWN LIMITATIONS

- Bridge is statistical: cannot identify any traveller
- Conversion merges purpose, residency and lodging choice
- No room inventory, events, fares or hotel prices
- Mixed 2022 vs. later flight grain; short history
- P10/P90 from only 7 months; assistant is rule-based

NEXT STEPS

- Pilot with DCT aviation and hotel planners on protected data
- Add purpose-of-travel, route-to-nationality, events, inventory, schedules and fares
- Production: access control, encryption, audit logs, versioned data, minimum-count suppression
- Retrain monthly, monitor drift, use bootstrapped intervals
- Optional governed LLM for the sentence box, returning only validated fields
- If shortlisted: rehearse the live demo end-to-end, run it live against new, unrehearsed routes in front of judges, and record a timed walkthrough video

Ask: a short DCT pilot to test assumptions, then replace inferred bridge weights with DCT's richer evidence.